

DPT Exercises

Exercise sheet 4

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Exercise 1. Let $(\Omega, \mathcal{A}, P)$ be a probability space and $X, Y : \Omega \rightarrow \mathbb{R}$, $Z_1, Z_2, Z_3 : \Omega \rightarrow \mathbb{R}^n$ random variables/vectors and $a, b \in \mathbb{R}$. Verify the following computation rules for variance and covariance by using their definitions and the properties of the expectation we have mentioned in the lecture:

$$\text{var}[aX + bY] = a^2 \text{var}[X] + b^2 \text{var}[Y] + 2ab \text{cov}[X, Y]$$

and

$$\text{cov}[aZ_1 + bZ_2, Z_3] = a \text{cov}[Z_1, Z_3] + b \text{cov}[Z_2, Z_3].$$

[Hint: You can use that for random vectors Z_1, Z_2 we have $\text{cov}[Z_1, Z_2] = \mathbb{E}[(Z_1 - \mathbb{E}[Z_1])^\top (Z_2 - \mathbb{E}[Z_2])]$, where $Z_1^\top$ is the transpose of Z_1 .

Solution: Since $\text{var}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2]$, using the linearity of the expectation gives

$$\begin{aligned} \text{var}[aX + bY] &= \mathbb{E}[(aX + bY - \mathbb{E}[aX + bY])^2] = \mathbb{E}[(aX + bY)^2] - (\mathbb{E}[aX + bY])^2 \\ &= (a^2 \mathbb{E}[X^2] + b^2 \mathbb{E}[Y^2] + 2ab \mathbb{E}[XY]) - (a^2 (\mathbb{E}[X])^2 + b^2 (\mathbb{E}[Y])^2 + 2ab \mathbb{E}[X] \mathbb{E}[Y]) \\ &= a^2 (\mathbb{E}[X^2] - (\mathbb{E}[X])^2) + b^2 (\mathbb{E}[Y^2] - (\mathbb{E}[Y])^2) + 2ab (\mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y]) \\ &= a^2 \text{var}[X] + b^2 \text{var}[Y] + 2ab \text{cov}[X, Y]. \end{aligned}$$

Since for general higher dimensional RVs, $\text{cov}[X, Y] = \mathbb{E}[(X - \mathbb{E}[X])^\top (Y - \mathbb{E}[Y])]$, by the linearity of expectation and the matrix multiplication being distributive, we obtain

$$\begin{aligned} \text{cov}[aZ_1 + bZ_2, Z_3] &= \mathbb{E}[(aZ_1 + bZ_2 - \mathbb{E}[aZ_1 + bZ_2])^\top (Z_3 - \mathbb{E}[Z_3])] \\ &= \mathbb{E}[a(Z_1 - \mathbb{E}[Z_1])^\top (Z_3 - \mathbb{E}[Z_3])] + \mathbb{E}[b(Z_2 - \mathbb{E}[Z_2])^\top (Z_3 - \mathbb{E}[Z_3])] \\ &= a \text{cov}[Z_1, Z_3] + b \text{cov}[Z_2, Z_3]. \end{aligned}$$

Exercise 2. Calculate the mean and variance for

- (i) the discrete RV $X : \Omega \rightarrow \mathbb{N}$ with pmf $\rho(n) = \frac{\lambda^n}{n!} e^{-\lambda}$ for a fixed $\lambda > 0$.
- (ii) the binomially distributed RV $X : \Omega \rightarrow \{0, \dots, n\}$, meaning X models the number of successes observed in $n \in \mathbb{N}$ i.i.d trials where each trial is Bernoulli distributed with probability of success $p \in (0, 1)$.

[Hint: Consider writing $X = \sum_{i=1}^n Y_i$ for $Y_i \stackrel{\text{iid}}{\sim} \text{Ber}(n, p)$ and use Exercise 1.]

Solution:

- (i) For the mean, we calculate

$$\mathbb{E}[X] = \sum_{n=0}^{\infty} n \cdot \frac{\lambda^n}{n!} e^{-\lambda} = \sum_{n=1}^{\infty} \frac{\lambda^n}{(n-1)!} e^{-\lambda} = \lambda e^{-\lambda} \sum_{n=1}^{\infty} \frac{\lambda^{n-1}}{(n-1)!}$$

and with the substitution $m = n - 1$, we obtain

$$= \lambda e^{-\lambda} \underbrace{\sum_{m=0}^{\infty} \frac{\lambda^m}{m!}}_{e^\lambda} = \lambda$$

For the variance, notice first that

$$\mathbb{E}[X(X-1)] = \sum_{n=0}^{\infty} \frac{n \cdot (n-1) \cdot \lambda^n}{n!} e^{-\lambda} = \sum_{n=2}^{\infty} \frac{\lambda^n}{(n-2)!} e^{-\lambda} = \lambda^2 e^{-\lambda} \sum_{m=0}^{\infty} \frac{\lambda^m}{m!} = \lambda^2$$

hence

$$\text{var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \mathbb{E}[X(X-1)] + \mathbb{E}[X] - (\mathbb{E}[X])^2 = \lambda.$$

(ii) First note that the Binomial-distributed RV can be expressed as

$$X = \sum_{i=1}^n Y_i$$

with Y_i i.i.d. copies of RV $Y : \Omega \rightarrow \{0, 1\}$ with $P(Y = 1) = p \in (0, 1)$, $P(Y = 0) = (1 - p)$ with expectation and variance given as

$$\mathbb{E}[Y] = 1 \cdot P(Y = 1) = p, \quad \text{var}[Y] = p - p^2 = p(1 - p).$$

Then, using that the expectation is linear and the Y_i s are independent, we get

$$\mathbb{E}[X] = \sum_{i=1}^n \mathbb{E}[Y_i] = n \cdot p$$

and

$$\text{var}[X] = \sum_{i=1}^n \text{var}[X_i] = n \cdot p \cdot (1 - p).$$

Exercise 3. Let X and Y be two jointly continuous random variables with joint pdf

$$p_{X,Y}(x, y) = \begin{cases} \frac{x^2}{4} + \frac{y^2}{4} + \frac{xy}{6} & \text{if } x \in [0, 1], y \in [0, 2], \\ 0 & \text{otherwise.} \end{cases}$$

For $y \in [0, 2]$, find

- (i) the conditional pdf of X given $Y = y$, i.e., the function $p_{X|Y}(x | y)$,
- (ii) the probability $P(X < \frac{1}{2} | Y = y)$.

Solution:

- (i) We first have to calculate the marginal pdf of Y , namely

$$p_Y(y) = \int_0^1 \left(\frac{x^2}{4} + \frac{y^2}{4} + \frac{xy}{6} \right) dx = \frac{3y^2 + y + 1}{12}, \quad \text{for } 0 \leq y \leq 2.$$

Thus for $y \in [0, 2]$, we have

$$p_{X|Y}(x | y) = \frac{p_{X,Y}(x, y)}{p_Y(y)} = \frac{\frac{3x^2+3y^2+2xy}{12}}{\frac{3y^2+y+1}{12}} = \frac{3x^2 + 3y^2 + 2xy}{3y^2 + y + 1}, \quad x \in [0, 1],$$

yielding overall

$$p_{X|Y}(x | y) = \begin{cases} \frac{3x^2+3y^2+2xy}{3y^2+y+1} & \text{if } x \in [0, 1], \\ 0 & \text{otherwise.} \end{cases}$$

(ii) We compute

$$\begin{aligned} P\left(X < \frac{1}{2} \mid Y = y\right) &= \int_0^{1/2} \frac{3x^2 + 3y^2 + 2xy}{3y^2 + y + 1} dx \\ &= \frac{1}{3y^2 + y + 1} [x^3 + yx^2 + 3y^2x]_0^{1/2} = \frac{\frac{1}{8} + \frac{y}{4} + \frac{3y^2}{2}}{3y^2 + y + 1}. \end{aligned}$$

Exercise 4. Let $(\Omega, \mathcal{A}, P)$ be a probability space and $X, Y : \Omega \rightarrow \mathbb{R}$ be random variables. Then show the following.

- (i) If X and Y are independent, they are uncorrelated.
- (ii) Give an example for X and Y being uncorrelated but not independent.
[Hint: Choose $Y = X^2$ for an appropriate choice of RV X . To show the non-independence, work with the cdfs.]

Solution:

- (i) Since $X \perp\!\!\!\perp Y$, we have $p_{X,Y}(x, y) = p_X(x)p_Y(y)$. Therefore

$$\mathbb{E}[XY] = \int_{\mathbb{R}^2} xy p_{X,Y}(x, y) d\lambda^2(x, y) = \int_{\mathbb{R}} x p_X(x) dx \int_{\mathbb{R}} y p_Y(y) dy = \mathbb{E}[X] \mathbb{E}[Y]$$

(the first integral with respect to the Lebesgue measure is only shown for completeness, it suffices if you directly write the Riemann integrals only) and thus

$$\text{cov}[X, Y] = \mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y] = 0.$$

Since the correlation coefficient has the covariance in the numerator, it is also zero.

- (ii) To show this, consider $X \sim \text{Unif}(-1, 1)$ and $Y = X^2$. We note that $\mathbb{E}[X] = (-1 + 1)/2 = 0$, $p_X(x) = \frac{1}{2}\chi_{[-1,1]}(x)$. The covariance is then

$$\text{cov}[X, Y] = \mathbb{E}[XY] - \underbrace{\mathbb{E}[X]}_{=0} \mathbb{E}[Y] = \int_{-1}^1 x^3 \cdot \frac{1}{2} dx = \frac{x^4}{8} \Big|_{-1}^1 = 0,$$

where we use that $\mathbb{E}[XY] = \mathbb{E}[X^3] = \int x^3 p_X(x) dx$. Hence, X and Y are uncorrelated. In order to establish the non-independence, we use that

$$X \perp\!\!\!\perp Y \Leftrightarrow F_X(x)F_Y(y) = F_{X,Y}(x, y)$$

and show that this factorization of cdfs is violated. For $y \in (0, 1]$, we have, since $p_X(x) = \frac{1}{2}$ for $x \in [-1, 1]$ and 0 otherwise,

$$F_Y(y) = P(Y \leq y) = P(X^2 \leq y) = P(X \in [-\sqrt{y}, \sqrt{y}]) = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2} dx = \sqrt{y}$$

and

$$F_X(x) = \int_{-1}^x \frac{1}{2} dx' = \frac{x}{2} + \frac{1}{2}.$$

On the other hand, notice that for $x \in [-1, 1]$, $y \in [0, 1]$ we have

$$\begin{aligned} F_{X,Y}(x, y) &= P(X \leq x, Y \leq y) = P(X \in [-1, x], X \in [-\sqrt{y}, \sqrt{y}]) = P(X \in [-\sqrt{y}, \min\{x, \sqrt{y}\}]) \\ &= \int_{-\sqrt{y}}^{\min\{x, \sqrt{y}\}} \frac{1}{2} dx' = \frac{1}{2} (\min\{x, \sqrt{y}\} + \sqrt{y}) \neq F_X(x)F_Y(y) = \frac{1}{2}(x\sqrt{y} + \sqrt{y}). \end{aligned}$$

Hence, X and Y are uncorrelated, but not independent. Can you visualize how the joint density of these variables looks like and what uncorrelatedness and non-independence mean visually?